CH SATYA VALLI

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Career Objective:

Automotive Software Testing Engineer with 4+ years of experience in embedded software verification and validation (V&V) across SW4–SW6 V-Model levels for safety-critical ECUs. Strong expertise in unit testing, integration testing, qualification testing, regression testing, and MC/DC structural code coverage using Embedded C. Hands-on experience in CAN/UDS protocol validation, on-target debugging using Trace32/JTAG, and requirement traceability using Polarion. Well-versed in ISO 26262, ASPICE processes, and automotive ECU validation with proven ability to ensure high-quality, compliant, and defect-free software delivery.

Professional Summary:

- Performed **Software Unit Testing, Integration Testing, Regression Testing, Qualification Testing, and Target-Based Testing** across **SW4–SW6 V-Model levels.**
- Strong expertise in **Grey-box and Black-box Testing** methodologies for **Automotive ECUs.**
- Reviewed **Software Detailed Design (SWDD)** documents to ensure **Requirement Clarity and Test Coverage Alignment.**
- Possess **1 year of hands-on experience in Embedded Firmware Analysis, Debugging, and Validation** for **Automotive ECUs.**
- Debugged embedded applications using **Trace32 and JTAG**, performing **Root Cause Analysis** at ECU level.
- Managed **End-to-End Defect Lifecycle** using **JIRA**, including reporting, tracking, and verification of fixes.
- Participated in **Code Walkthroughs and Peer Reviews** to improve **Software Quality and Defect Prevention.**
- Ensured **MC/DC Structural Code Coverage**, including **Statement and Branch Coverage Compliance.**
- Executed **Software Integration and Qualification Testing** for automotive ECUs.
- Developed and executed **Unit Test Cases using VectorCAST** based on **Functional and Design Requirements.**
- Compiled **Embedded Source Code** using appropriate compilers and analyzed **Software Interfaces and Dependencies.**
- Performed debugging using **Breakpoints, Probe Assertions, and Log Analysis Techniques.**
- Generated detailed **Test Summary, Coverage, and Validation Reports**, and actively participated in **Weekly Project Review Meetings** to discuss risks and progress.
- Possess strong **Interpersonal Skills**, self-motivated, **Result-Oriented**, and committed to **Continuous Learning and Quality Improvement.**

Professional Experience:

- Working as a **Software Test Engineer - Automotive |at BorgWarner** From May 2023 to Till date.
- Worked as a **Junior Software Engineer at People Tech Group** From May 2022 – April 2023.

Educational Qualification:

- Bachelor of Engineering – Computer Science Kakinada Institute of Engineering and Technology .

Technical Skills :

- **Programming Languages:** C, Embedded C (Low-Level Driver Interaction), Python
- **Operating Systems:** Windows, Linux
- **Microcontrollers:** 8051, ARM, PIC16F877A
- **Software Testing:** Functional Testing, System Testing, Integration Testing, Regression Testing, Qualification Testing
- **Test Development:** Test Plan Design, Test Case Development, Test Scenario Creation, Validation Procedures
- **Automation:** Python Scripting, Regression Automation, Test Framework Development
- **Testing & Debugging Tools:** TESSY, VectorCAST, Trace32, CANoe, JTAG
- **DevOps & ALM Tools:** JIRA, Jenkins (CI), Bitbucket (Source Control), Polarion
- **Embedded Systems:** ARM Microcontrollers, Real-Time Operating Systems (RTOS), Communication Protocols (CAN, UDS)
- **Peripheral Drivers:** ADC, PWM, SPI, UART, I2C, Timers, Interrupts

Mini Projects (Embedded C):

Developed mini projects including UART-based bit banking communication system, temperature controller, calculator application, and LCD & LED interfacing. Implemented serial communication using UART protocol,

real-time temperature monitoring using ADC, and display/control using LCD and LEDs. These projects enhanced hands-on experience in embedded C programming, peripheral interfacing, and real-time system validation.

Worked PROJECTS:

Project 1: eTurbo – Porsche 911

Tools: TESSY, Polarion, Bitbucket, Trace32, JIRA, Doxygen | Compiler: Green Hills

- Developed unit test cases based on SWDD requirements.
- Generated MC/DC, statement, and branch coverage reports.
- Tracked and resolved defects using JIRA.

Project Description:

The eTurbo integrates a mechanical turbocharger with an electric motor, providing electrified boost assistance for superior response. The system integrates embedded software, sensors, actuators, and control units (ECUs) to deliver real-time boost control under varying driving conditions while meeting strict automotive safety and performance standards.

Roles and Responsibilities:

- Analyzed **system and software requirements** for the eTurbo control unit.
- Reviewed Software Detailed Design (SWDD) documents.- Debugged embedded applications using Trace32 and JTAG.
- Managed defect tracking in JIRA.
- Conducted code walkthroughs and peer reviews.
- Ensured MC/DC structural coverage.
- Designed and executed **embedded software test cases** based on functional and safety requirements.
- Performed **unit, integration, and system testing** on ECU-level software.
- Performed **qualification and integration testing** for automotive ECUs using **BorgWarner flash software**
- Executed test cases and validated functionality using **CANoe**
- Analyzed test results, identified issues, and reported defects
- Uploaded test results, logs, and evidence to **Polarion** for traceability and documentation

Project 2: e-Heater ECU Validation & Automation

Project Description:

The project involved validation and automation testing of an automotive e-Heater Electronic Control Unit (ECU) used for thermal management in electric and hybrid vehicles. The scope covered qualification, integration, and regression testing to ensure reliable ECU functionality, robust CAN/LIN communication, and compliance with system requirements. Automation frameworks were leveraged to improve test efficiency, regression coverage, and reporting.

Roles and Responsibilities:

- Performed **qualification, integration, and regression testing** of the automotive e-Heater ECU across multiple test cycles.
- Validated **CAN and communication** using **Vector CANoe** along with **Vector VN hardware**.
- Conducted **on-target debugging** using **Trace32 (Lauterbach)** via **JTAG/SWD** interfaces to analyze firmware behavior.
- Executed **Python-based automation frameworks** for regression testing and **power-cycle test scenarios.**
- Automated test execution to improve coverage, repeatability, and overall testing efficiency.
- Generated detailed **test reports** and uploaded **test results to Polarion** for requirements traceability and defect tracking.
- Collaborated with development and system teams to analyze failures and support issue resolution.

Project 3: Advanced Driver Assistance System (ADAS)

Developed functional and robustness test cases.

Created integration test cases and structural coverage reports.

Performed validation of safety systems.

Performed sensor data validation and fault-injection testing for ADAS modules.

Project Description:

Advanced Driver Assistance Systems (ADAS) are safety-critical embedded systems designed to enhance vehicle safety and driving comfort by assisting drivers in perception, decision- making, and control. ADAS integrates data from multiple sensors such as cameras, radar, ultrasonic sensors, and ECUs to enable features like Adaptive Cruise Control (ACC), Lane Keeping Assist (LKA), Automatic Emergency Braking (AEB), Blind Spot Detection, and Parking Assist. The system operates under strict real-time and functional safety constraints in compliance with automotive standards such as **ISO 26262** and **ASPICE**.

Roles and Responsibilities:

- Reviewed **system, software, and safety requirements** for ACU functionality.
- Analyzed **system, software, and safety requirements** for ADAS functionalities.
- Designed and executed **embedded test cases** for sensor processing, control logic, and decision algorithms.
- Performed **unit, integration** across multiple ECUs.
- Supported **functional safety (ISO 26262)** and **ASPICE compliance** activities.